

Foundations of Cybersecurity/Applied Cryptography

19 February 2024

Exercise n. 1 [12] [All]

With reference to a Public Key Infrastructure (PKI)

1. Illustrate the main fields in a digital certificate.
2. Explain the difference between invalid and revoked certificate.
3. Argue whether the following certificate formats are secure or not
 - a. $A, \text{pubK}_A, V_A, \text{Sign}_A(A||\text{pubK}_A||V_A)$
 - b. $A, \text{pubK}_A, V_A, \text{Sign}_{CA}(A||\text{pubK}_A||V_A)$
 - c. $A, \text{pubK}_A, V_A, \text{Sign}_B(A||\text{pubK}_A||V_A)$
 - d. $[A, \text{pubK}_A, V_A, \text{Sign}_B(A||\text{pubK}_A||V_A)], [B, \text{pubK}_B, \text{Sign}_{CA}(B||\text{pubK}_B||V_B||CA=true)]$

Exercise n. 2 [10] [All]

Let $h: \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function that is second-preimage and collision resistant. Let $h': \{0, 1\}^* \rightarrow \{0, 1\}^{n+1}$ be the hash function given by the rule

$$h'(x) = \begin{cases} 0||x & x \in \{0,1\}^n \\ 1||h(x) & \text{otherwise} \end{cases}$$

Prove that h' is not preimage resistant but still second-preimage and collision resistant.

Exercise n. 3 [8] [2022-23]

Alice and Bob use the Diffie-Hellman key establishment protocol. Let Y_A and Y_B be the respective public keys. To avoid the MitM attack, Alice and Bob resort to the trusted third-party Trent that acts as certification authority. The trust relationships concretize themselves in the pre-shared symmetric keys K_A e K_B , respectively. Assuming that clocks are not synchronized, design a protocol that allows us to achieve the following beliefs:

A believes Y_B is Bob's public key B believes Y_A is Alice's public key

Exercise n. 3 [8] [earlier]

Find and patch vulnerabilities in the following function, if any.

```
unsigned int do_stuff(unsigned int a, unsigned int b){
    if(a*b > UINT_MAX){
        /*Handle Error*/
    }
    return a*b;
}
```

Solution

EXERCISE N.1

Question 1 and 2: see theory.

Question 3.

- Insecure as self-signed.
- Secure.
- Insecure: B has not been delegated.
- Secure: B has been delegated.

EXERCISE N.2

The modified hash function h' is not preimage resistant, since for any hash value y of the form $0||x$, a preimage is x . Therefore, we can find a preimage for at least one half of all possible hash values.

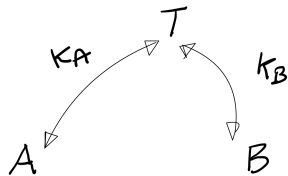
Now we prove that h' inherits second preimage and collision resistance from h . We show that if we can find a collision or a second preimage for h' , then we can easily do so for h . Suppose $\exists x_0 \neq x_1$: $h'(x_0) = h'(x_1)$.

There are two cases:

- First bit of $h'(x_0)$ is 0. This is impossible as it implies that $x_0 = x_1$.
- First bit of $h'(x_0)$ is 1. Then $h(x_0) = h(x_1)$, but this is a contradiction, as h is collision resistant.

EXERCISE N.3 [2022-23]

The protocol that allows Alice to get the belief.



$A \rightarrow T: A, B, n_A$

$T \rightarrow B: A, B, n_A$

$B \rightarrow T: \{A, B, n_A, \gamma_B\}_{K_B}$

$T \rightarrow A: \{A, B, n_A, \gamma_B\}_{K_A}$

The protocol that allows Bob to get to the belief is the dual.

EXERCISE N.3 [EARLIER]

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2 February 2024

The problem is wraparound. The error condition is written in the wrong way as, if $a*b$ wraps-around, the `if` clause will never be true. The following patch must be applied.

```
unsigned int do_stuff(unsigned int a, unsigned int b){
    if(b!=0 && a > UINT_MAX/b){
        /*Handle Error*/
    }
    return a*b;
}
```

WARNING! The `if`-clause cannot be written in the inverse order as $(a>UINT_MAX/b)$ raises an error if b is zero